

ulms-highgrade-pulm-mets-tp53-smarc b1-hla-a2-w9t4

Standard of care options — approved and guideline-endorsed strategies

Generated 2026-08-25

Standard of care options

The treatment strategies that are standard for this patient's situation, meaning a regulator approved them for a population that includes this patient or a major academic or clinical-society guideline carries them. This is one of the case's two therapeutic tables and frequently carries its primary recommendation; the experimental ranking is the other, and neither narrows the other.

Assessed: 16 standard options. **Consider now:** 4. **Behind an open gate:** 3. **Already received:** 0.

Sixteen standard options were screened for this systemic-therapy-naïve metastatic uterine leiomyosarcoma, against NCCN Uterine Neoplasms v2.2026, NCCN Soft Tissue Sarcoma v1.2025, the ESMO-EURACAN-GENTURIS sarcoma guideline, and current FDA labels. Four are open now, and the first is the regimen already planned: doxorubicin + trabectedin with trabectedin maintenance (LMS-04), preferred by NCCN for uterine LMS and the board's first choice on four of five slates. Its trial doubled progression-free survival and moved median survival from 24 to 33 months, though no uterine-only effect size has been published and the regimen's cost, including 28% febrile neutropenia, has not yet been discussed with her. Gemcitabine + docetaxel, doxorubicin + ifosfamide, and single-agent doxorubicin are the endorsed alternatives.

Nothing systemic is spent. Her only prior treatment is the 2024 hysterectomy, so the whole first-line landscape is open, and that is why sequencing dominates this table: off-protocol gemcitabine permanently closes the abemaciclib trial seat (NCT06498648), which LMS-04 preserves, and every doxorubicin cycle draws down two later trial ceilings (360 mg/m² for peposertib, exactly six LMS-04 cycles; 450 mg for in-vivo lung perfusion). The induction cycle count is a before-cycle-1 decision.

Three label routes sit behind open gates: pembrolizumab and dostarlimab on TMB and MSI/MMR that were never measured, larotrectinib on an NTRK fusion never tested on tissue. One door is already closed by results: ER/PR under 1% forecloses endocrine therapy. Metastasectomy and SBRT wait at restaging; their favourable series describe countable nodules, not her current infiltrative bilateral pattern.

This report adds the standard options alongside the experimental ranking. It does not narrow, reorder, or argue against anything ranked there.

Options to consider now

Standard options this patient is eligible for, with an endorsement that covers their situation, and no unresolved gate in front of them.

1. Doxorubicin + trabectedin, then trabectedin maintenance (LMS-04 regimen)

Priority: essential **Modality:** Systemic chemotherapy **Line:** First line **Intent:** life-prolonging intent

What makes it standard.

- NCCN Preferred, first-line systemic therapy for uterine sarcoma: 'Doxorubicin, Trabectedin (for LMS)' (covers this patient · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: First-line systemic therapy for uterine sarcoma; the doxorubicin + trabectedin entry is annotated for

leiomyosarcoma (Guide 7 of the patient-guide derivative).

Fit to this patient. Conditionally eligible Systemic-therapy naive with this exact regimen already planned, so nothing in the record argues against it; the conditions are the profile's own named open items: baseline echocardiogram before doxorubicin, hepatic and renal panels before trabectedin, an altitude-referenced pulmonary and oximetry baseline (she lives at about 8,500 ft), and confirmation of the assumed ECOG 1.

- blocker: No baseline LVEF/echocardiogram on file; the profile names it mandatory before doxorubicin
- blocker: No hepatic or renal labs on file; trabectedin carries hepatotoxicity
- blocker: Pulmonary reserve and oximetry not yet assessed against high-altitude residence
- blocker: ECOG 1 is an unverified intake assumption

The regimen the consulting sarcoma oncologist has already planned, and the board's first choice on four of five slates: in LMS-04, adding trabectedin to doxorubicin with a trabectedin maintenance phase roughly doubled median PFS (HR 0.41, 95% CI 0.29-0.58) and extended median OS to 33 vs 24 months (adjusted HR 0.65, 95% CI 0.44-0.95) in first-line advanced leiomyosarcoma, and NCCN now carries it as a preferred first-line option for uterine LMS. Two caveats carry weight: uterine primaries were a 45% stratification stratum and no uterine-only effect size has been published, so the headline numbers are trial-wide rather than uterine-specific; and the toxicity price (80% grade 3-4 neutropenia, 28% febrile neutropenia) has not been put to her, since every preference on file is an intake default she never stated. The named baseline workup (echo, hepatic panel, altitude-referenced oximetry, a documented ECOG) belongs before cycle 1.

Key evidence.

- LMS-04 (primary report) · randomised open-label phase 3 vs doxorubicin alone · n=150 · mPFS 12.2 mo (95% CI 10.1-15.6) vs 6.2 mo (4.1-7.1); adjusted HR 0.41 (95% CI 0.29-0.58)
- LMS-04 (final overall survival) · final OS analysis of the same phase 3 · n=150 · mOS 33 mo (95% CI 26-48) vs 24 mo (19-31); adjusted HR 0.65 (95% CI 0.44-0.95)

Toxicities that would change the decision. Grade 3-4 neutropenia 80%, thrombocytopenia 47%, febrile neutropenia 28% (vs 13%, 0% and 9% with doxorubicin alone); one treatment-related death in the trial; Anthracycline cardiotoxicity; cumulative dose planning matters from cycle 1; Trabectedin hepatotoxicity and rare rhabdomyolysis; central line required

Alongside the targeted options. May foreclose a targeted option. Contains no gemcitabine, so it preserves the seat on the abemaciclib + gem/doc trial (NCT06498648, Experimental rank 3), which off-protocol gemcitabine would permanently close. Each cycle also draws down two anthracycline ceilings: peposertib (NCT05711615) requires cumulative doxorubicin of 360 mg/m² or less, exactly six LMS-04 cycles at 60 mg/m², and in-vivo lung perfusion (NCT02811523) excludes more than 450 mg cumulative. The induction cycle count is a before-cycle-1 decision; a five-cycle plan is a choice made up front, not at cycle six.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; pmid:35835135; pmid:39231341; nct:NCT02997358

2. Gemcitabine + docetaxel

Priority: high **Modality:** Systemic chemotherapy **Line:** First line **Intent:** life-prolonging intent

What makes it standard.

- NCCN Preferred, first-line systemic therapy for uterine sarcoma: 'Docetaxel, Gemcitabine' (covers this patient · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: First-line systemic therapy for uterine sarcoma, all histologies (Guide 7 of the patient-guide derivative).

Fit to this patient. Likely eligible No contraindication in the record; no labs of any kind are on file, and gemcitabine's occasional severe pulmonary toxicity deserves weight in a patient with infiltrative bilateral lung disease and an unassessed high-altitude respiratory baseline.

The other NCCN-preferred first-line option for uterine LMS and the historical US default, with real single-arm first-line activity in this exact histology (ORR 35.8%, 90% CI 23.5-49.6). The randomised footing is sobering: in GeDDiS it was not superior to doxorubicin and the PFS hazard ratio of 1.28 (95% CI 0.99-1.65) trended worse, which is part of why the board's slates centred on the doxorubicin-trabectedin backbone instead. Its main sequencing property is what it closes: any off-protocol gemcitabine permanently forecloses the abemaciclib + gem/doc trial seat.

Key evidence.

- GeDDiS · randomised phase 3 vs doxorubicin, first line · n=257 · 24-week PFS 46.4% (95% CI 37.5-54.8) vs 46.3% with doxorubicin; HR for PFS 1.28 (95% CI 0.99-1.65), direction favouring doxorubicin
- GOG first-line uLMS phase 2 · single-arm phase 2, first line · n=42 · ORR 35.8% (90% CI 23.5-49.6); mPFS 4.4 mo

Toxicities that would change the decision. Myelosuppression; more fatigue than doxorubicin in GeDDiS, less mucositis and febrile neutropenia; Gemcitabine pulmonary toxicity (a G4 event occurred in the uLMS phase 2), weighed against her infiltrative bilateral lung disease and altitude baseline; Docetaxel neuropathy, oedema and hypersensitivity

Alongside the targeted options. May foreclose a targeted option. Off-protocol gemcitabine permanently closes NCT06498648 (abemaciclib + gem/doc, Experimental rank 3), which admits treatment-naive patients; the trial is the protocolised way to receive this same backbone. LMS-04 contains no gemcitabine and leaves that seat open, an asymmetry that belongs in the first-line conversation. Choosing gem/doc off protocol spends none of the anthracycline ceilings the doxorubicin-based options draw down.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; pmid:28882536; pmid:18534250

3. Doxorubicin + ifosfamide (with mesna)

Priority: high **Modality:** Systemic chemotherapy **Line:** First line **Intent:** life-prolonging intent

What makes it standard.

- NCCN Preferred, first-line systemic therapy for uterine sarcoma: 'Doxorubicin, Ifosfamide' (covers this patient · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: First-line systemic therapy for uterine sarcoma (Guide 7 of the patient-guide derivative).

Fit to this patient. Conditionally eligible Same anthracycline preconditions as the planned regimen (baseline echo, organ-function labs, altitude-referenced pulmonary baseline, confirmed ECOG), plus renal function for ifosfamide; none of these values is on file.

- blocker: No baseline LVEF/echocardiogram on file (mandatory before doxorubicin)
- blocker: No renal panel on file; ifosfamide is nephrotoxic
- blocker: ECOG 1 is an unverified intake assumption

The intensified first-line doublet for a patient in whom response, not just control, is the goal: EORTC 62012 doubled the response rate (26% vs 14%) and improved PFS, but overall survival did not move (HR 0.83, 95.5% CI 0.67-1.03) and febrile neutropenia reached 46% despite pegfilgrastim. NCCN carries it as a preferred first-line option for uterine sarcoma, so it stays a live alternative where cytoreduction toward local therapy is the aim. The response-for-toxicity trade is exactly the kind of decision the record shows has not yet been discussed with her.

Key evidence.

- EORTC 62012 · randomised phase 3 vs doxorubicin alone, first line · n=455 · ORR 26% vs 14%; mPFS 7.4 vs 4.6 mo (HR 0.74, 95% CI 0.60-0.90); mOS 14.3 vs 12.8 mo, HR 0.83 (95.5% CI 0.67-1.03), p=0.076, primary endpoint not met; febrile neutropenia 46% despite pegfilgrastim

Toxicities that would change the decision. Febrile neutropenia 46% vs 13% with doxorubicin alone, despite growth-factor support; Ifosfamide neurotoxicity (encephalopathy) and nephrotoxicity; Anthracycline cardiotoxicity; higher per-cycle doxorubicin dose (75 mg/m²) reaches the cumulative ceilings sooner than LMS-04

Alongside the targeted options. May foreclose a targeted option. Draws down the same two anthracycline ceilings as LMS-04, and faster: at 75 mg/m² per cycle, the 360 mg/m² cap for peposertib (NCT05711615) is crossed inside five cycles and the 450 mg cumulative exclusion for in-vivo lung perfusion (NCT02811523) follows shortly after. Contains no gemcitabine, so the abemaciclib + gem/doc trial seat stays open.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; pmid:24618336; nct:NCT00061984

4. Doxorubicin, single agent

Priority: high **Modality:** Systemic chemotherapy **Line:** First line **Intent:** life-prolonging intent

What makes it standard.

- NCCN Preferred, first-line systemic therapy for uterine sarcoma: 'Doxorubicin' (covers this patient · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: First-line systemic therapy for uterine sarcoma (Guide 7 of the patient-guide derivative).
- FDA on-label approval (covers this patient · current label via openFDA, retrieved August 2026)
 - covers: Doxorubicin hydrochloride injection is indicated for the treatment of metastatic soft tissue sarcoma (among other tumours).

Fit to this patient. Conditionally eligible The least toxic of the preferred first-line options, but the anthracycline preconditions are unchanged: no baseline echocardiogram, labs, or documented ECOG is on file, and her high-altitude respiratory baseline is unmeasured.

- blocker: No baseline LVEF/echocardiogram on file (mandatory before doxorubicin)
- blocker: ECOG 1 is an unverified intake assumption

Single-agent doxorubicin is the on-label, guideline-preferred backbone that both LMS-04 and EORTC 62012 improved on in PFS or response without ever beating it decisively on survival until LMS-04's final analysis, and it remains the gentlest of the preferred first-line options (febrile neutropenia around 9-13% vs 28% and 46% for the combinations). For a patient whose efficacy-toxicity weighting is an unstated intake default, it is the fallback the combinations have to justify themselves against. NCCN's uterine sarcoma list carries it as preferred, and its label covers metastatic soft tissue sarcoma outright.

Key evidence.

- EORTC 62012 (doxorubicin control arm) · control arm of a randomised phase 3 · n=228 · mPFS 4.6 mo, mOS 12.8 mo (95.5% CI 10.5-14.3), ORR 14%; the reference performance every first-line comparison above is measured against

Toxicities that would change the decision. Grade 3-4 neutropenia 13% and febrile neutropenia 9-13% in trial control arms, far below either combination; Cumulative cardiotoxicity; the same ceiling arithmetic as the combinations applies

Alongside the targeted options. May foreclose a targeted option. Same anthracycline-ceiling arithmetic as the combinations: cumulative doxorubicin above 360 mg/m² excludes peposertib (NCT05711615) and above 450 mg excludes in-vivo lung perfusion (NCT02811523). No gemcitabine, so the abemaciclib + gem/doc seat stays open.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; fda:doxorubicin hydrochloride injection label (openFDA); pmid:24618336

Options at the next decision point

Standard for a line this patient has not reached yet, or for a disease state that has not arrived. Listed now so the sequence is visible in advance.

5. Trabectedin, second line after anthracycline (uterine leiomyosarcoma)

Priority: high **Modality:** Systemic chemotherapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (partial population match · approval 2015-10; current label via openFDA, retrieved August 2026)
 - covers: YONDELIS is indicated for the treatment of adult patients with unresectable or metastatic liposarcoma or leiomyosarcoma who received a prior anthracycline-containing regimen.
- NCCN Second-line, preferred therapy for leiomyosarcomas: 'Trabectedin (Yondelis) - for leiomyosarcomas' (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Second-line chemotherapy for uterine sarcoma; trabectedin is the preferred entry for leiomyosarcoma (Guide 10 of the patient-guide derivative).

Fit to this patient. Conditionally eligible Histology matches the label exactly, but she is anthracycline-naive, so the labelled population (prior anthracycline-containing regimen) is one she has not yet entered; hepatic function, which gates trabectedin dosing, is also not on file.

- blocker: Label and guideline position both assume a prior anthracycline-containing regimen; she has had no systemic therapy
- blocker: No hepatic panel on file

The on-label second-line standard for uterine LMS after an anthracycline (PFS HR 0.57, 95% CI 0.41-0.81 in the uterine subgroup), and NCCN's preferred second-line entry for leiomyosarcoma. Its availability at second line depends directly on the first-line choice: LMS-04 spends trabectedin in induction and maintenance, so progression on that regimen leaves this option already used, whereas a gemcitabine-based first line keeps trabectedin intact as the labelled next step. That contingency is worth naming before cycle 1, not after progression.

Key evidence.

- ET743-SAR-3007 · randomised phase 3 vs dacarbazine, after anthracycline · n=518 · mPFS 4.2 vs 1.5 mo, HR 0.55, p<0.001; interim OS HR 0.87 (mOS 12.4 vs 12.9 mo)
- ET743-SAR-3007 uterine LMS subgroup · post hoc subgroup of the same phase 3 · n=232 · mPFS 4.0 vs 1.5 mo, HR 0.57 (95% CI 0.41-0.81); ORR 11%; OS HR 0.89 (95% CI 0.65-1.24)

Toxicities that would change the decision. Grade 3-4 transaminitis (ALT 34% in the uLMS subset) and neutropenia (36%); treatment-related deaths 2.1% in the trial's trabectedin arm; Requires central venous access and dexamethasone premedication

Last verified. 2026-08-25

References. fda:YONDELIS label (openFDA); guideline:NCCN Uterine Neoplasms v2.2026; pmid:26371143; pmid:28651804; nct:NCT01343277

6. Pulmonary metastasectomy after systemic control

Priority: medium **Modality:** Surgery **Line:** Not line-indexed **Intent:** disease-control intent

What makes it standard.

- NCCN Option for limited metastatic disease: surgical removal of metastases (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Metastatic uterine sarcoma with few metastatic tumours: removing them with surgery, or destroying them with EBRT, SBRT, or image-guided ablation, may be an option (uterine sarcoma chapter of the patient-guide derivative).
- ESMO Standard option for resectable pulmonary metastases (ESMO-EURACAN-GENTURIS soft tissue sarcoma CPG) (partial population match · 2021 (Ann Oncol; current published version))
 - covers: Complete surgical excision of resectable lung metastases in advanced soft tissue sarcoma, in selected patients with limited, resectable disease.

Fit to this patient. Conditionally eligible The consulting sarcoma oncologist explicitly kept local therapy on the table after systemic control, but the current radiology describes bilateral, infiltrative, possibly inflammatory disease, which is not the countable-nodule phenotype the surgical series enrolled; feasibility is contingent on the pattern

after induction, and her pulmonary reserve at 8,500 ft is unmeasured.

- blocker: Infiltrative bilateral pattern; the favourable metastasectomy series enrolled complete resections of countable nodules and excluded this phenotype by construction
- blocker: Pulmonary function and oximetry not assessed against high-altitude residence
- blocker: Feasibility depends on response to first-line systemic therapy

Resection of limited lung metastases is the one standard route to long-term survival this disease offers (5-year OS 55.4% in the uterine-primary registry; median OS 70 months in selected LMS series), which is why the treating oncologist kept it on the table after systemic control. The honesty constraint is that those figures belong to complete resections of countable nodules, and her current imaging shows the opposite phenotype: bilateral, infiltrative, possibly inflammatory. Whether induction converts her into the population the series describe is a question for restaging, and the stakes of that reassessment are high because this is the closest thing to a curative-intent move on either table.

Key evidence.

- Japanese multicentre registry, uterine primaries · retrospective multicentre registry · n=319 · 5-year OS 55.4% after metastasectomy for uterine primaries; the figure belongs to complete resections of countable nodules, with disease-free interval under 12 months carrying an elevated adjusted hazard of death
- Repeated metastasectomy in LMS · retrospective single-institution cohort · n=82 · median OS 70 months in selected patients undergoing repeated resection (vs 24 months for non-LMS histologies, p=0.049)

Toxicities that would change the decision. Operative morbidity of bilateral or staged thoracic surgery, against an unmeasured high-altitude pulmonary baseline; Loss of lung parenchyma in a patient with chronic hypoxic adaptation at 8,500 ft

Alongside the targeted options. Competes for the same line of therapy. The in-vivo lung perfusion trial (NCT02811523, Experimental rank 5) and pulmonary suffusion are metastasectomy-plus-perfusion packages occupying this same post-induction local-therapy slot; IVLP excludes cumulative doxorubicin above 450 mg, so the induction plan chosen now decides whether the surgical route can still be the trial version or only the standard one.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; pmid:34303806; pmid:37063124; pmid:21867989

7. Stereotactic body radiotherapy (SBRT) to lung metastases

Priority: medium **Modality:** Radiotherapy **Line:** Not line-indexed **Intent:** disease-control intent

What makes it standard.

- NCCN Option for limited metastatic disease: SBRT among the listed local therapies (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Metastatic uterine sarcoma with few metastatic tumours: local options include image-guided ablation, EBRT, and stereotactic body radiation therapy (uterine sarcoma chapter of the patient-guide derivative).

Fit to this patient. Conditionally eligible Ablative radiotherapy assumes a countable set of targets; her infiltrative

bilateral pattern does not currently offer that, and the pulmonary-reserve question that governs how much lung can safely be treated is unmeasured against her altitude baseline.

- blocker: Infiltrative bilateral pattern is not the discrete-target phenotype the SBRT series treated
- blocker: Pulmonary function and oximetry not assessed against high-altitude residence
- blocker: Feasibility depends on response to first-line systemic therapy

The non-surgical route to local control of limited lung disease, carried by NCCN's uterine sarcoma pathway for patients with few metastatic tumours, with high local-control figures in sarcoma-specific series (24-month local failure 7.4%, 95% CI 4.0-13.9). Both supporting series are retrospective and selected, so the reported absence of severe toxicity is an artefact-prone estimate rather than a guarantee, and her infiltrative bilateral disease is not yet an SBRT geometry. Like metastasectomy, this becomes a real decision at restaging after induction.

Key evidence.

- MSK series · retrospective single-institution cohort · n=66 · 24-month cumulative local failure 7.4% (95% CI 4.0-13.9); no grade 3+ toxicity reported
- Cleveland Clinic series · retrospective single-institution cohort · n=50 · 3-year local control 88%; 76% needed no new or changed systemic therapy after SBRT

Toxicities that would change the decision. Radiation pneumonitis and fibrosis: the zero grade 3+ figure across the two retrospective series is retrospective ascertainment in selected patients, not a demonstrated safety margin, and her reduced-reserve, high-altitude lungs are exactly where that distinction matters

Alongside the targeted options. Competes for the same line of therapy. Occupies the same post-induction local-consolidation slot as metastasectomy and the surgical-perfusion trial packages (IVLP, suffusion); ablating index lesions may also change eligibility for lung-directed trial procedures, so the local-therapy modality is best chosen once, at restaging, with all of these on the table.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; pmid:37532104; pmid:36914598

8. Pazopanib

Priority: medium **Modality:** Targeted therapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (partial population match · approval 2012-04; current label via openFDA, retrieved August 2026)
 - covers: VOTRIENT is indicated for the treatment of adults with advanced soft tissue sarcoma who have received prior chemotherapy. Limitation of use: efficacy not demonstrated in adipocytic STS or GIST.
- NCCN Second-line, listed among recommended therapies: 'Pazopanib (Votrient)' (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Second-line chemotherapy and targeted therapy for uterine sarcoma (Guide 10 of the patient-guide derivative; the derivative's phrasing scopes the list beyond the trabectedin-preferred leiomyosarcoma entry).

Fit to this patient. Conditionally eligible The label population requires prior chemotherapy, which she has not yet had; hepatic function, the main pazopanib gate, is not on file.

- blocker: Label requires prior chemotherapy; she is systemic-therapy naive
- blocker: No hepatic panel on file; pazopanib carries boxed hepatotoxicity

An on-label oral option after chemotherapy: PALETTE tripled median PFS (4.6 vs 1.6 months, HR 0.31, 95% CI 0.24-0.40) without moving overall survival, and the uterine-specific footing is a retrospective analysis with an 11% response rate. It belongs in the later-line conversation as a chemotherapy-free interval rather than a survival play. Not applicable now because the label population starts after prior chemotherapy.

Key evidence.

- PALETTE · randomised double-blind placebo-controlled phase 3, after prior chemotherapy · n=369 · mPFS 4.6 vs 1.6 mo, HR 0.31 (95% CI 0.24-0.40); OS not improved, HR 0.86 (95% CI 0.67-1.11)
- EORTC 62043/62072 uterine sarcoma analysis · retrospective analysis of two prospective trials · n=44 · ORR 11% (95% CI 3.8-24.6); mPFS 3.0 mo (95% CI 2.5-4.7)

Toxicities that would change the decision. Boxed hepatotoxicity warning; LFT monitoring against a baseline that is not yet on file; Hypertension, fatigue, diarrhoea; pneumothorax has been reported with TKIs in patients with pleural-adjacent lung metastases

Last verified. 2026-08-25

References. fda:VOTRIENT label (openFDA); guideline:NCCN Uterine Neoplasms v2.2026; pmid:22595799; pmid:27012429; nct:NCT00753688

9. Gemcitabine + dacarbazine

Priority: medium **Modality:** Systemic chemotherapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- NCCN Second-line, listed among recommended therapies: 'Gemcitabine, Dacarbazine' (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Second-line chemotherapy for uterine sarcoma (Guide 10 of the patient-guide derivative; the derivative's phrasing scopes the list beyond the trabectedin-preferred leiomyosarcoma entry).

Fit to this patient. Likely eligible No contraindication in the record; positioned at second line by guideline carriage rather than by any label constraint, and the gemcitabine pulmonary caveat noted on the gem/doc row applies here too.

A guideline-listed later-line doublet with an unusually strong randomised phase 2 behind it: against dacarbazine alone, adding gemcitabine doubled median overall survival (16.8 vs 8.2 months, HR 0.56, 95% CI 0.36-0.90) in an LMS-enriched population, at modest toxicity. It is a genuine second-line contender rather than a completeness entry, and its gemcitabine content carries the same trial-foreclosure consequence as gem/doc.

Key evidence.

- Spanish GEIS randomised phase 2 · randomised phase 2 vs dacarbazine alone · n=113 · 3-month PFS rate 56% vs 37%; PFS HR 0.58 (95% CI 0.39-0.86); mOS 16.8 vs 8.2 mo, OS HR 0.56 (95% CI 0.36-0.90)

Toxicities that would change the decision. Generally well tolerated; granulocytopenia the main serious event, febrile neutropenia uncommon; Gemcitabine pulmonary toxicity, weighed against infiltrative bilateral lung disease

Alongside the targeted options. May foreclose a targeted option. Contains gemcitabine, so the same foreclosure as gem/doc: any off-protocol gemcitabine closes NCT06498648 (abemaciclib + gem/doc) if that seat is still open when this option comes up.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; pmid:21606430

10. Eribulin

Priority: low **Modality:** Systemic chemotherapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (liposarcoma, not leiomyosarcoma) (different population · approval 2016-01; current label via openFDA, retrieved August 2026)
 - covers: HALAVEN is indicated for the treatment of patients with unresectable or metastatic liposarcoma who have received a prior anthracycline-containing regimen.
- NCCN Second-line, listed among recommended therapies: 'Eribulin (Halaven)' (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Second-line chemotherapy for uterine sarcoma (Guide 10 of the patient-guide derivative; the derivative's phrasing scopes the list beyond the trabectedin-preferred leiomyosarcoma entry).

Fit to this patient. Conditionally eligible Physically nothing bars it, but the trial and label position it after an anthracycline, a line she has not reached, and the histology on the label is liposarcoma rather than leiomyosarcoma.

- blocker: Label and trial population assume a prior anthracycline; she is systemic-therapy naive
- blocker: FDA indication is liposarcoma; the LMS subgroup showed no OS advantage

Carried for completeness at the later lines: the phase 3 overall survival benefit (HR 0.77, 95% CI 0.62-0.95) was concentrated in liposarcoma, the FDA approval is liposarcoma-only, and the leiomyosarcoma subgroup did not show an OS advantage. NCCN's uterine sarcoma second-line list still includes it, so it stays available as a later-line option, but the histology-specific evidence puts it behind trabectedin, pazopanib and gemcitabine-dacarbazine in this patient's sequence.

Key evidence.

- Eribulin vs dacarbazine phase 3 · randomised open-label phase 3, after anthracycline · n=452 · mOS 13.5 vs 11.5 mo, HR 0.77 (95% CI 0.62-0.95) overall; the benefit concentrated in liposarcoma, with no OS advantage in the leiomyosarcoma subgroup
- Liposarcoma subgroup analysis · preplanned subgroup analysis of the same phase 3 · n=143 · OS benefit driven by the liposarcoma stratum; basis of the liposarcoma-restricted FDA approval

Toxicities that would change the decision. Grade 3+ adverse events 67% (vs 56% with dacarbazine); neutropenia the characteristic high-grade event; Peripheral neuropathy with cumulative dosing

Last verified. 2026-08-25

References. fda:HALAVEN label (openFDA); guideline:NCCN Uterine Neoplasms v2.2026; pmid:26874885; pmid:28854066; nct:NCT01327885

11. Temozolomide

Priority: medium **Modality:** Systemic chemotherapy **Line:** Third line or later **Intent:** disease-control intent

What makes it standard.

- NCCN Second-line, listed among recommended therapies: 'Temozolomide (Temodar)' (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Second-line chemotherapy for uterine sarcoma (Guide 10 of the patient-guide derivative; the derivative's phrasing scopes the list beyond the trabectedin-preferred leiomyosarcoma entry).

Fit to this patient. Likely eligible Oral, gentle by chemotherapy standards, and unconstrained by any label gate in this setting; its position is later-line by guideline carriage and by the modest activity data.

An oral later-line option whose modest single-agent activity clusters in leiomyosarcoma (both responses in the supporting phase 2 were LMS), carried on NCCN's uterine sarcoma second-line list and notable for how little toxicity it asks. Its main strategic weight in this case is indirect: the temozolomide-containing combinations on the experimental table are where the board put its interest, and off-protocol single-agent use could complicate entry to them.

Key evidence.

- Temozolomide phase 2 in STS · single-arm phase 2 · n=25 · ORR 8% overall, with both responses in leiomyosarcoma; mPFS 3.9 mo in the LMS subset (95% CI 1.9-21.9); no grade 4 events

Toxicities that would change the decision. Mild myelosuppression and nausea; no grade 4 events in the supporting phase 2

Alongside the targeted options. May foreclose a targeted option. Single-agent temozolomide off protocol may affect eligibility for the ranked temozolomide combinations, olaparib + temozolomide (Experimental rank 2) and tuvusertib + temozolomide; protocol prior-exposure rules should be checked before it is used alone.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026; pmid:14584078

Standard options behind an open gate

Endorsed for this patient's situation, but gated on a result that is missing, pending, or below threshold. The gate has to close before the option is real.

12. Pembrolizumab (tumor-agnostic, TMB-high or MSI-high/dMMR)

Priority: medium **Modality:** Immunotherapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (tumor-agnostic, accelerated for TMB-H) (partial population match · current label via openFDA, retrieved August 2026)
 - covers: Unresectable or metastatic MSI-H or dMMR solid tumors, or TMB-H (≥ 10 mutations/megabase) solid tumors, as determined by an FDA-authorized test, that have progressed following prior treatment and with no satisfactory alternative treatment options.
- NCCN Second-line, therapies that may be used in certain circumstances: 'Immunotherapy (for TMB-H tumors): Pembrolizumab (Keytruda)' (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Second-line therapy for uterine sarcoma, restricted to TMB-H tumours (Guide 10 of the patient-guide derivative).

Fit to this patient. Conditionally eligible Two conditions stand between her and this label: the qualifying biomarker (TMB or MSI/MMR) has never been measured on tissue, and the tumor-agnostic indications require progression after prior treatment, a line she has not reached.

- blocker: TMB and MSI/MMR never measured; plasma assay did not report either and is unreliable at low tumor fraction
- blocker: Label requires progression following prior treatment; she is systemic-therapy naive

Biomarker gate. TMB (tissue) or MSI/MMR (tissue) TMB ≥ 10 mutations/megabase, or MSI-H/dMMR, by an FDA-authorized test (never measured)

The only route by which checkpoint blockade is standard for her is biomarker-selected and later-line: the tumor-agnostic labels require a TMB or MSI/MMR result that has never been generated (tissue testing is already part of the case's shared workup order) plus progression on prior therapy. The unselected evidence in this histology is flatly negative (0 of 10 LMS in SARC028; a dedicated uLMS study stopped for futility at 0 of 12), so this row is a gate to close cheaply, not a therapy to reach for. MSI-H is rare in uLMS, which keeps expectations low even with the test ordered.

Key evidence.

- KEYNOTE-158 (prospective TMB analysis) · prospective biomarker analysis of a multicohort single-arm phase 2 · $n=790$ · ORR 29% in TMB-high vs 6% in non-TMB-high tumours; basis of the tumour-agnostic accelerated approval
- SARC028 (LMS cohort) · two-cohort single-arm phase 2 of pembrolizumab in sarcoma · $n=80$ · 0 of 10 leiomyosarcoma responses; a dedicated uLMS checkpoint study separately stopped for futility at 0 of 12. Unselected checkpoint blockade is inactive in this histology

Toxicities that would change the decision. Immune-related pneumonitis is the toxicity that would matter most against infiltrative bilateral lung disease and a high-altitude respiratory baseline; Endocrinopathies, colitis, hepatitis per the PD-1 class

Alongside the targeted options. May foreclose a targeted option. Prior PD-1 exposure can exclude patients from PD-1-based trial seats, including the ivonescimab study the board weighed (Experimental rank 6); if the gate ever

opens, the trial-versus-label sequencing is worth checking before dosing.

Last verified. 2026-08-25

References. fda:KEYTRUDA label (openFDA); guideline:NCCN Uterine Neoplasms v2.2026; pmid:32919526; pmid:28988646; pmid:31682550

13. Dostarlimab (tumor-agnostic, dMMR)

Priority: low **Modality:** Immunotherapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (tumor-agnostic, accelerated) (partial population match · current label via openFDA, retrieved August 2026)
 - covers: JEMPERLI as a single agent for adult patients with dMMR recurrent or advanced solid tumors, as determined by an FDA-approved test, that have progressed on or following prior treatment and who have no satisfactory alternative treatment options.

Fit to this patient. Conditionally eligible Same two conditions as pembrolizumab: MMR status has never been measured on tissue, and the label requires progression on prior treatment.

- blocker: MMR/MSI never measured on tissue
- blocker: Label requires progression on or following prior treatment; she is systemic-therapy naive

Biomarker gate. MMR (tissue IHC) / MSI dMMR by an FDA-approved test (never measured)

The second tumor-agnostic PD-1 route behind the identical unmet gate: dMMR status has never been measured, dMMR is rare in uterine LMS, and the label starts after progression on prior therapy. Carried so the report shows both label routes through the same door; closing the MMR gate is one tissue IHC on material that already exists.

Key evidence.

- GARNET (dMMR solid-tumour cohorts) · phase 1, single-group, multicentre; interim analysis · n=327 · ORR 44.0% (95% CI 38.6-49.6); median duration of response not reached; 72.2% of responses lasted 12+ months

Toxicities that would change the decision. PD-1 class immune-related events; pneumonitis carries extra weight against her lung disease and altitude baseline

Alongside the targeted options. May foreclose a targeted option. As with pembrolizumab, prior PD-1 exposure can close PD-1-based trial seats such as the ivonescimab study; the two label routes share one gate and one sequencing caveat.

Last verified. 2026-08-25

References. fda:JEMPERLI label (openFDA); pmid:37917058; nct:NCT02715284

14. Larotrectinib (NTRK fusion-positive tumors)

Priority: low **Modality:** Targeted therapy **Line:** Any line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (tumor-agnostic) (partial population match · current label via openFDA, retrieved August 2026)
 - covers: Adult and pediatric patients with solid tumors that have an NTRK gene fusion without a known acquired resistance mutation, are metastatic or where surgical resection is likely to result in severe morbidity, and have no satisfactory alternative treatments or that have progressed following treatment.
- NCCN First-line, therapies used in certain circumstances: 'TKI therapy for NTRK gene fusion-positive tumors: Larotrectinib (Vitrakvi)' (partial population match · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: First-line systemic therapy for uterine sarcoma, restricted to NTRK fusion-positive tumours (Guide 7 of the patient-guide derivative; entrectinib and repotrectinib are listed alongside).

Fit to this patient. Conditionally eligible Everything turns on a fusion that has never been tested at decision-relevant resolution: the plasma panel's non-detection at an unsupplied tumor fraction is not exclusionary, and no RNA-based or tissue fusion assay has been run.

- blocker: NTRK fusion status unmeasured on tissue; plasma non-detection at unknown tumor fraction does not exclude a fusion

Biomarker gate. NTRK1/2/3 gene fusion NTRK gene fusion without a known acquired resistance mutation, by an FDA-approved or equivalent assay (RNA-based fusion panel preferred) (status unknown)

A tumor-agnostic approval with dramatic response rates (ORR 79%, 95% CI 72-85) that NCCN carries even at first line for fusion-positive uterine sarcoma, sitting behind a biomarker that is vanishingly rare in leiomyosarcoma and has never been properly tested here. The plasma panel's silence does not close the question, because fusion detection in plasma is insensitive at low tumor fraction; an RNA-based fusion assay on the existing tissue would settle it. Low priority reflects the prior probability, not the drug.

Key evidence.

- Pooled larotrectinib analysis · pooled analysis of three phase 1/2 trials · n=159 · ORR 79% (95% CI 72-85), 16% complete responses; grade 3-4 treatment-related events each at 3% or below

Toxicities that would change the decision. Generally mild: transaminase rises, dizziness, fatigue; withdrawal pain on stopping

Last verified. 2026-08-25

References. fda:VITRAKVI label (openFDA); guideline:NCCN Uterine Neoplasms v2.2026; pmid:32105622; pmid:29466156

Assessed and set aside

Screened against this patient and found not to apply. Listed for the audit trail.

15. Gemcitabine + docetaxel + bevacizumab

Priority: low **Modality:** Systemic chemotherapy **Line:** First line **Intent:** disease-control intent

What makes it standard.

- NCCN Bevacizumab carried for a different sarcoma histology: 'Solitary fibrous tumor: Bevacizumab and temozolomide' (Soft Tissue Sarcoma guideline) (different population · v1.2025 (May 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Bevacizumab-containing therapy appears in the soft tissue sarcoma guideline for solitary fibrous tumor; the current uterine-sarcoma systemic lists (first- and second-line, v2.2026 derivative) do not carry bevacizumab in any combination.

Fit to this patient. Likely eligible No medical bar in the record; the reason this row is set aside is evidential and regulatory, not eligibility.

Assessed because the routing worklist carried it, and set aside on its own randomised evidence: GOG-0250 tested exactly this regimen in exactly this population, chemotherapy-naive metastatic uterine LMS, and stopped early for futility with the hazard ratios pointing the wrong way. Bevacizumab holds no FDA sarcoma indication, and the current uterine-sarcoma guideline lists do not carry it, so the plain gem/doc row above is the live version of this backbone.

Key evidence.

- GOG-0250 · randomised double-blind placebo-controlled phase 3, first line · n=107 · Bevacizumab added nothing: PFS HR 1.12 (p=0.58; mPFS 6.2 mo placebo vs 4.2 mo bevacizumab), OS HR 1.07 (p=0.81); accrual stopped early for futility

Toxicities that would change the decision. Bevacizumab added grade 3 hypertension (8% vs 0%) and a fistula in GOG-0250, with no efficacy return

Last verified. 2026-08-25

References. guideline:NCCN Soft Tissue Sarcoma v1.2025; pmid:25713428

16. Aromatase inhibition / anti-estrogen therapy

Priority: low **Modality:** Endocrine therapy **Line:** Any line **Intent:** disease-control intent

What makes it standard.

- NCCN Anti-estrogen therapy for hormone receptor-positive uterine sarcoma; Guide 8 lists aromatase inhibitors as preferred for low-grade ESS and low-risk adenosarcoma (different population · v2.2026 (November 2025; verified via the public NCCN patient-guide derivative, subscriber text not reachable this session))
 - covers: Uterine sarcoma that is hormone receptor-positive: anti-estrogen therapy may be recommended; the preferred anti-estrogen list (aromatase inhibitors, with or without GnRH analog) is written for low-grade ESS and low-risk adenosarcoma.

Fit to this patient. Likely ineligible Her tumor is ER and PR <1% by IHC, a confirmed decision-relevant negative; the activity of aromatase inhibition in uLMS is specific to ER-positive disease, so the receptor result closes this door rather than leaving it ajar.

- blocker: ER <1% and PR <1% by IHC, confirmed; the endorsement requires hormone receptor-positive disease

Biomarker gate. ER/PR expression (IHC) hormone receptor-positive (threshold not met)

Included so the absence of endocrine therapy reads as a decision, not an oversight: anti-estrogen therapy is the guideline's carve-out for hormone receptor-positive uterine sarcoma, and her confirmed ER/PR <1% takes her out of that population. This is the one standard-of-care door her existing biomarker results have already closed.

Last verified. 2026-08-25

References. guideline:NCCN Uterine Neoplasms v2.2026